Hugo Marquerie 21/10/2025

Prop ideal radical iff cociente reducido

Proposición 1. *Sea $I \subsetneq R$ un ideal, entonces*

$$I \text{ es un ideal radical} \iff {}^{R}\!/_{I} \text{ es un anillo reducido.}$$

***Demostración*:** Veamos la equivalencia directamente:

$$\begin{aligned} I \text{ es radical} &\iff I = \sqrt{I} = \{a \in R : \exists n \in \mathbb{N} : a^n \in I\} \\ &\iff \forall a \in R : \big((\exists n \in \mathbb{N} : a^n \in I) \implies a \in I \big) \\ &\iff \forall \overline{a} \in {}^{R}\!/_{I} : \big(\big(\exists n \in \mathbb{N} : \overline{a}^n = \overline{0}\big) \implies \overline{a} = \overline{0} \big) \\ &\iff {}^{R}\!/_{I} \text{ es un anillo reducido.} \end{aligned}$$

■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.